

Training Configuration and Summary:

Dataset used: rssi_odom_dataset-ORIGINAL.parquet

LSTM Multi-Step Training Report

Report Generated: 2025-05-07 16:03:56

Training Start: 2025-05-07 15:55:18

Training End: 2025-05-07 16:02:01

Training Duration: 6.72 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: elu

Optimizer: adam

Dropout Rate: 0.20

MAE: 1.7972

MSE: 5.9275

RMSE: 2.4346

R2 Score: -0.0873

Multi-Step Forecast:

Forecast Steps: 40

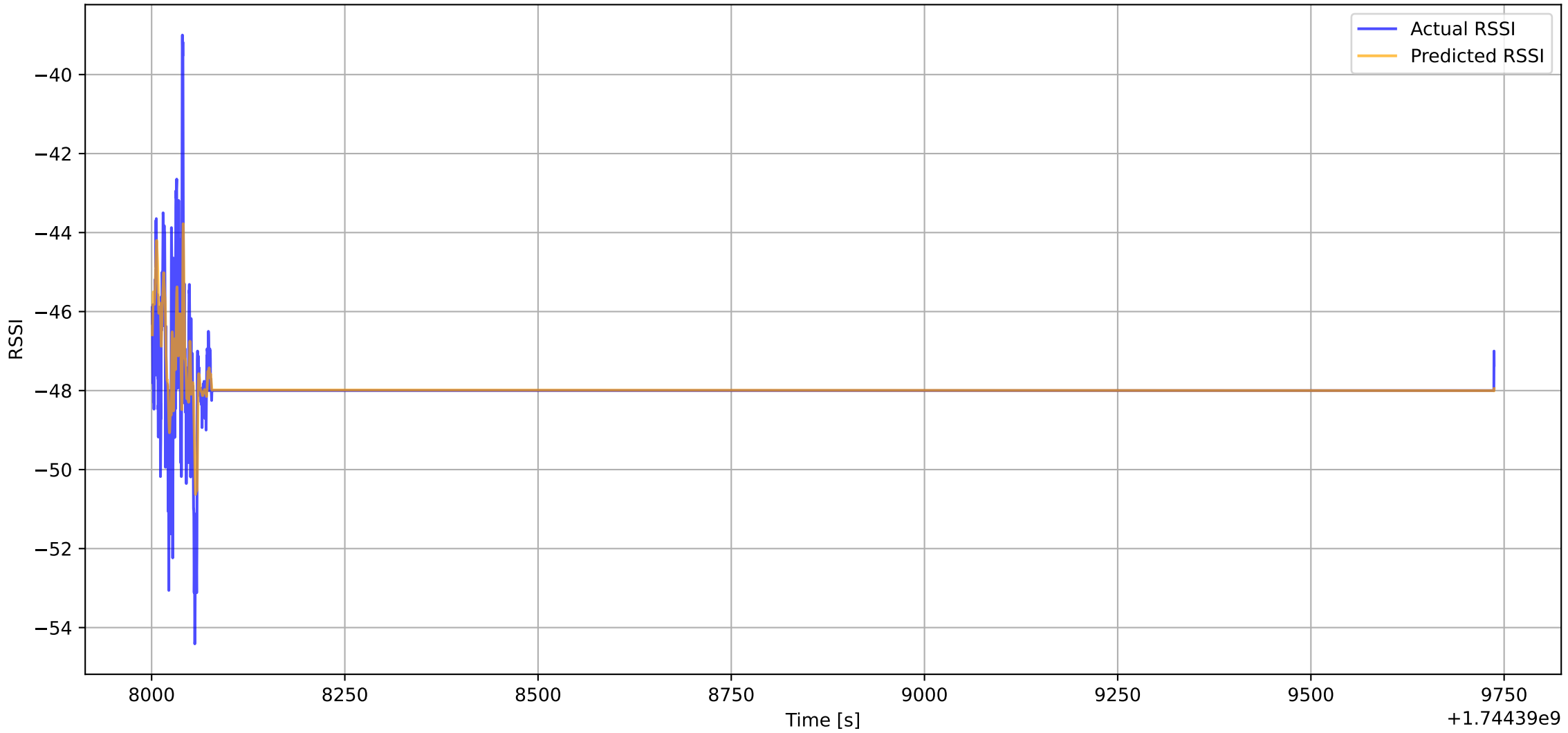
MAE: 0.0554

MSE: 0.0057

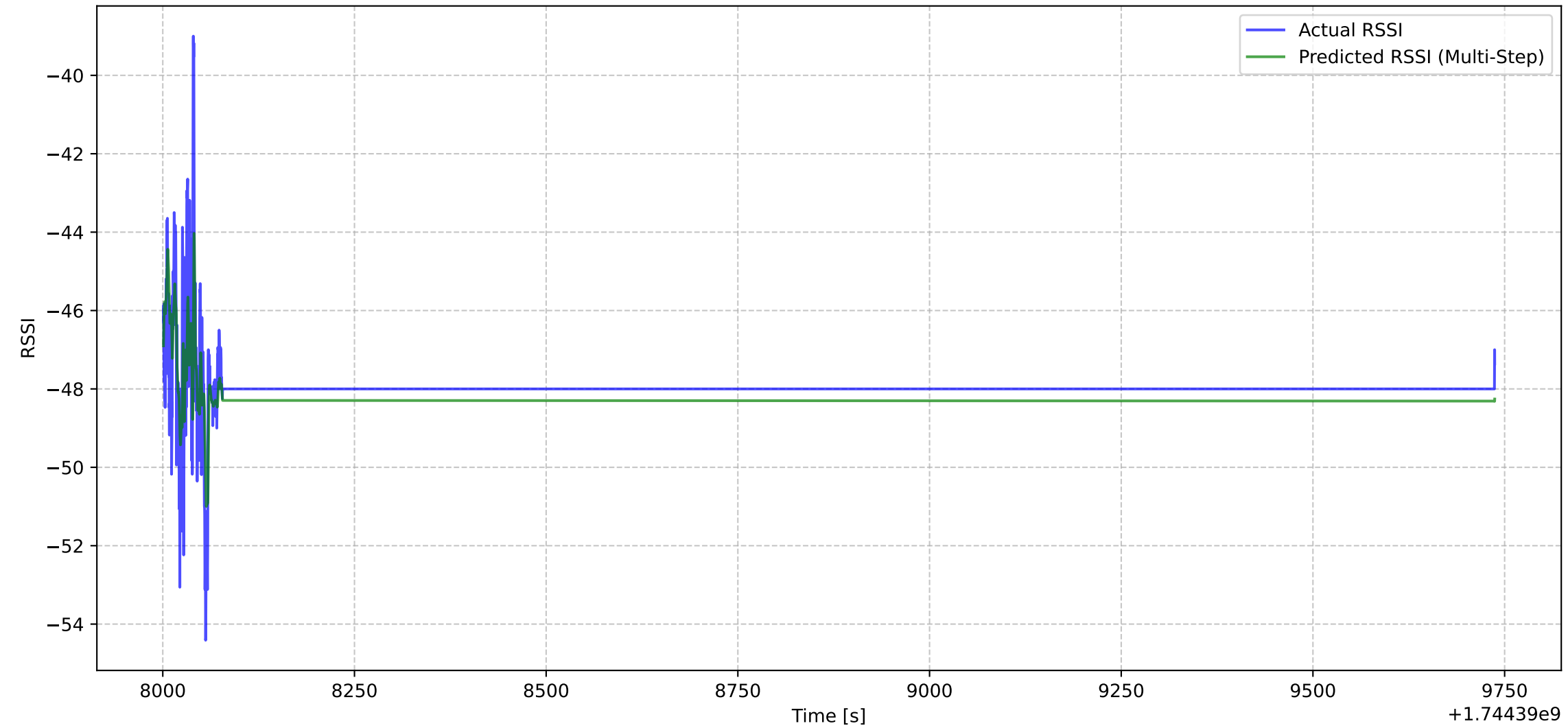
RMSE: 0.0754

R2 Score: 0.3683

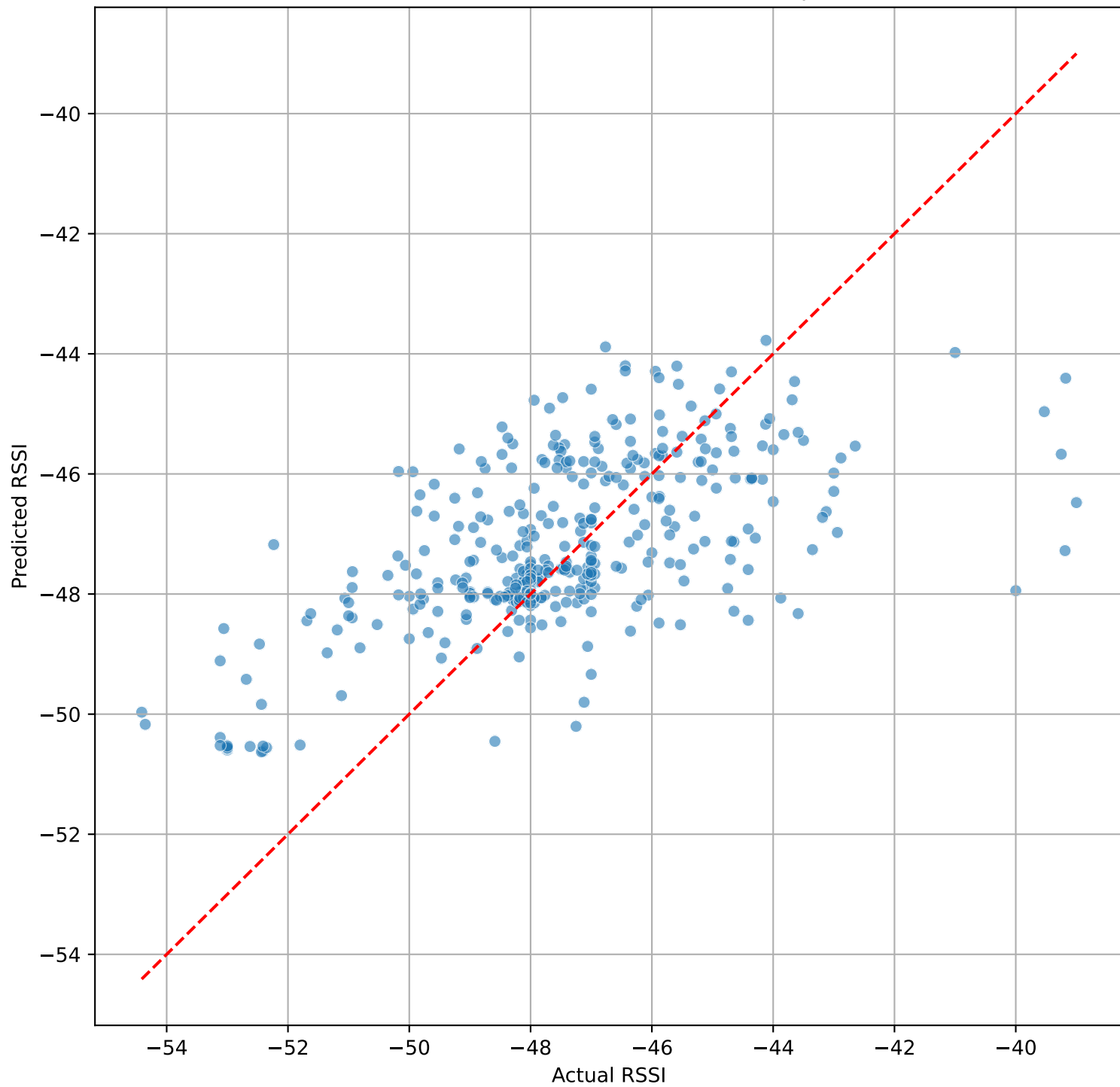
Actual vs Predicted RSSI (First Step)



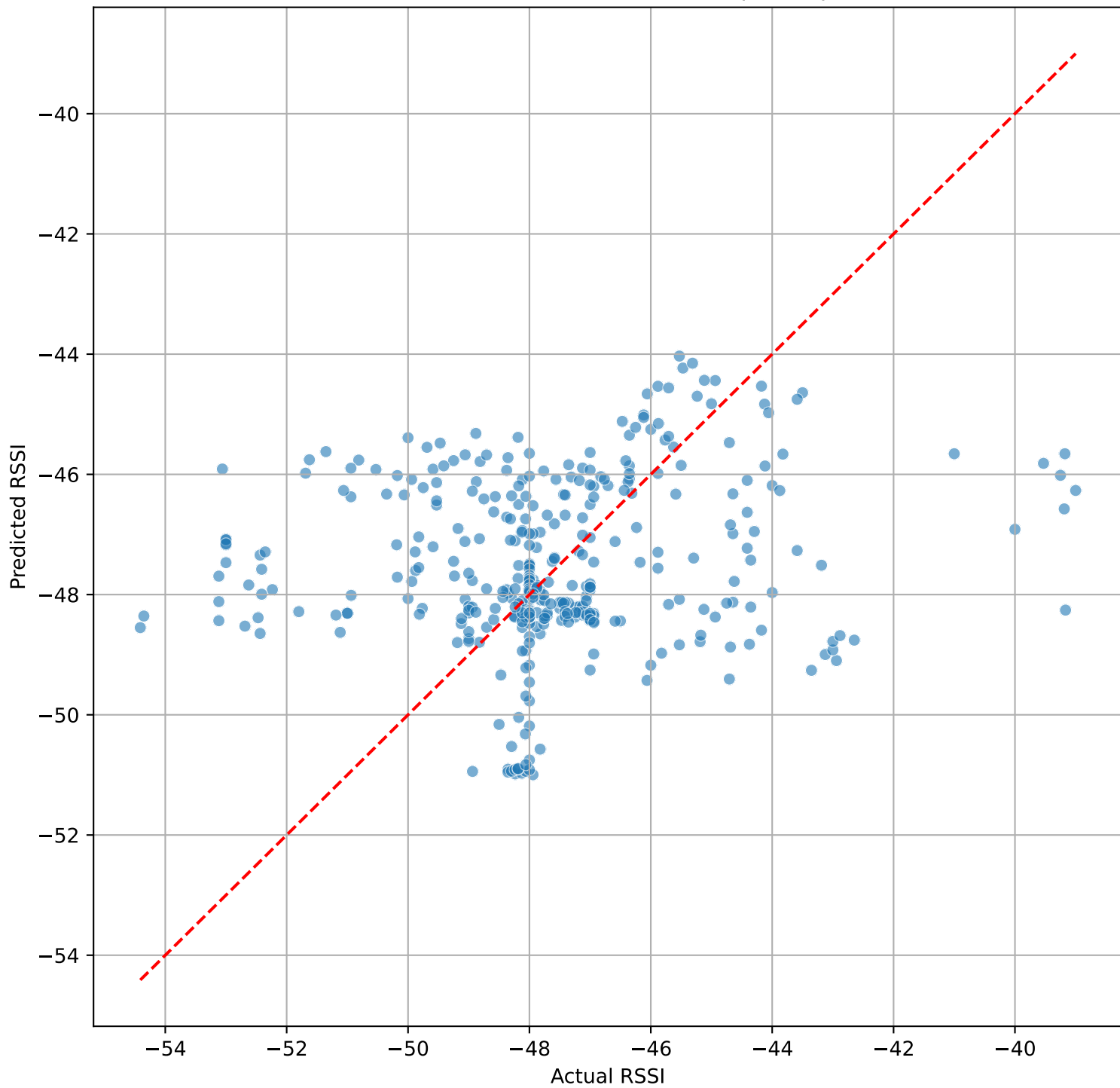
Actual vs Predicted RSSI (Multi-Step (step 40))



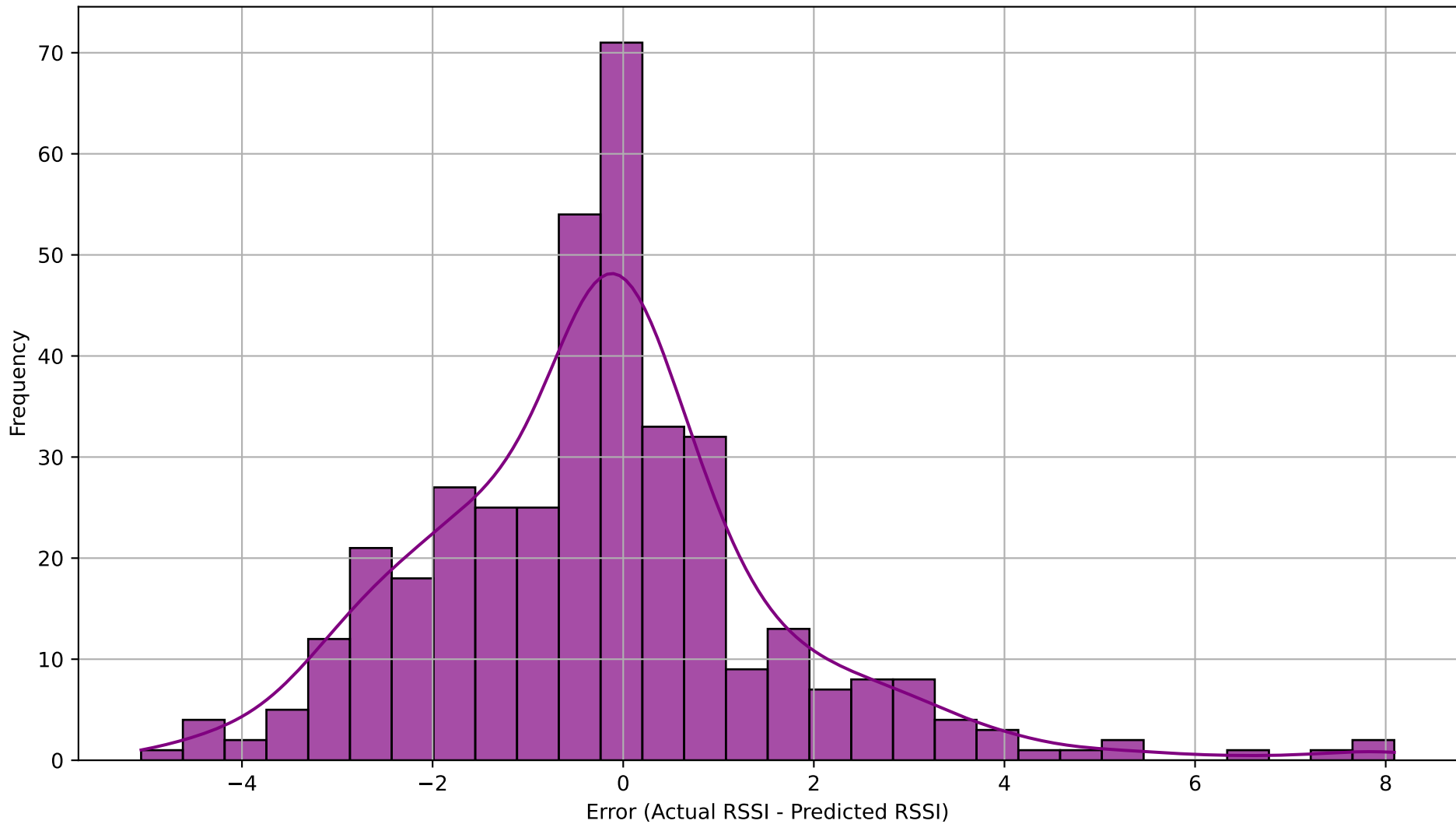
Actual vs Predicted RSSI (First Step)



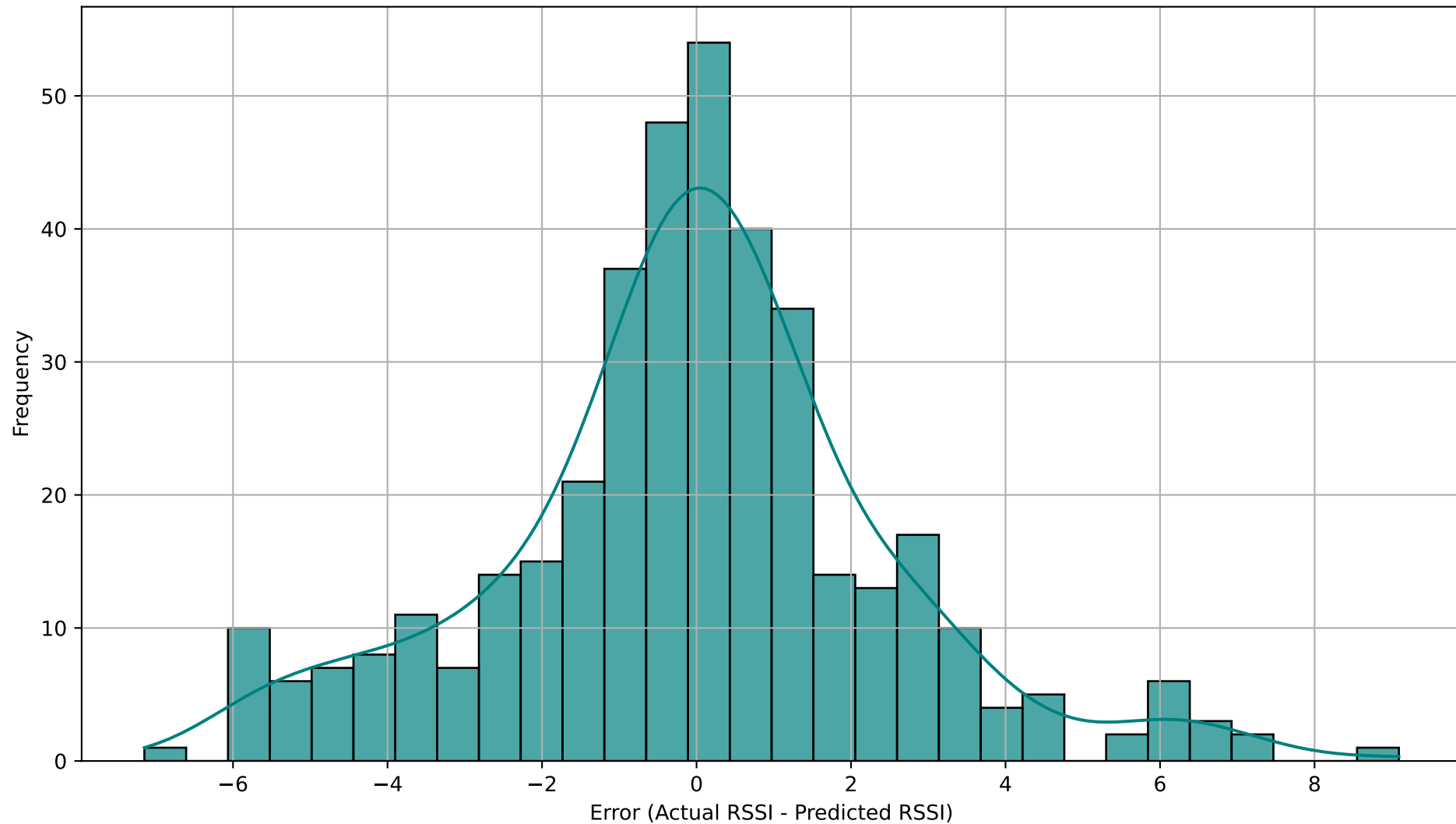
Actual vs Predicted RSSI (Last Step - step 40)



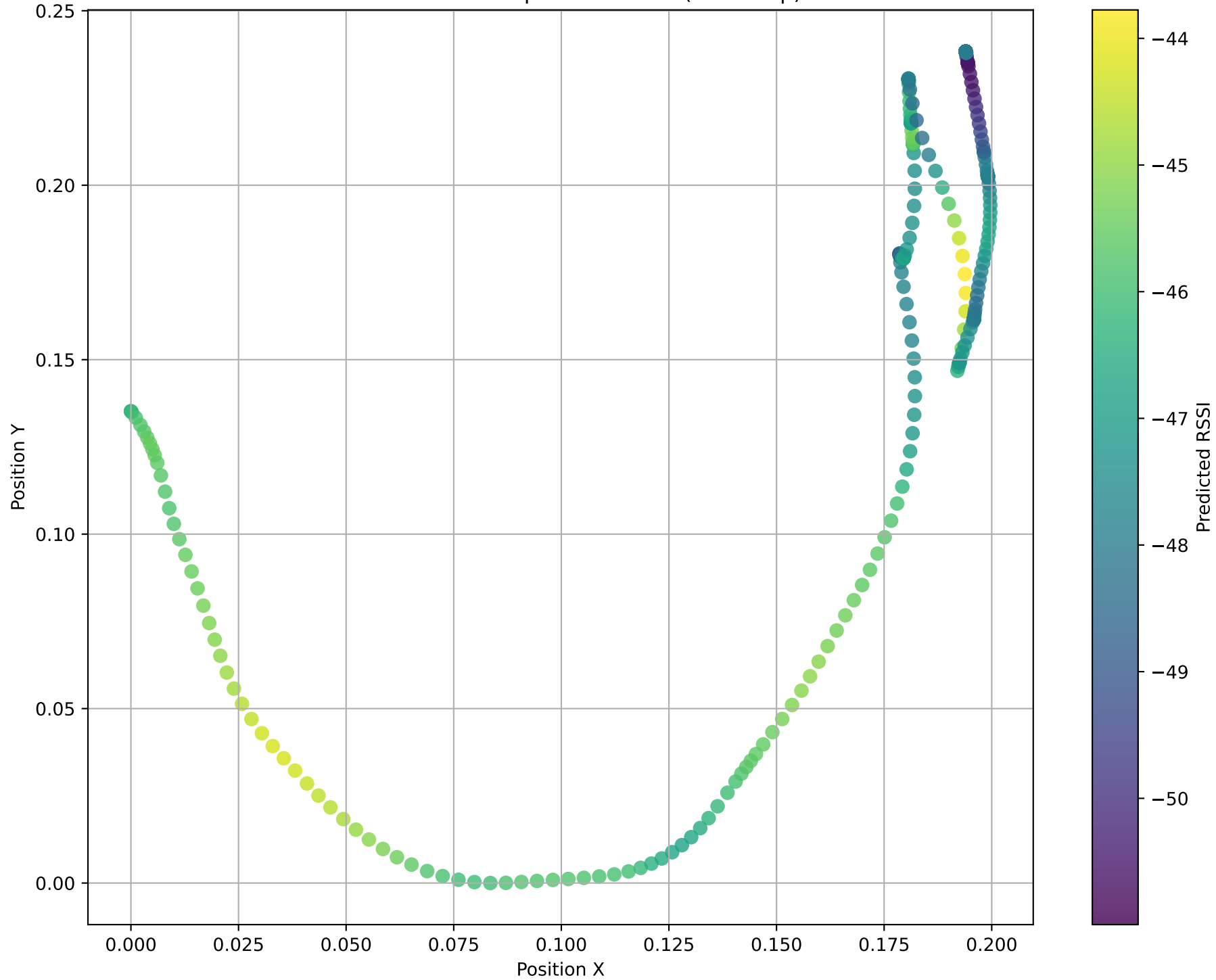
Distribution of Prediction Errors (First Step)



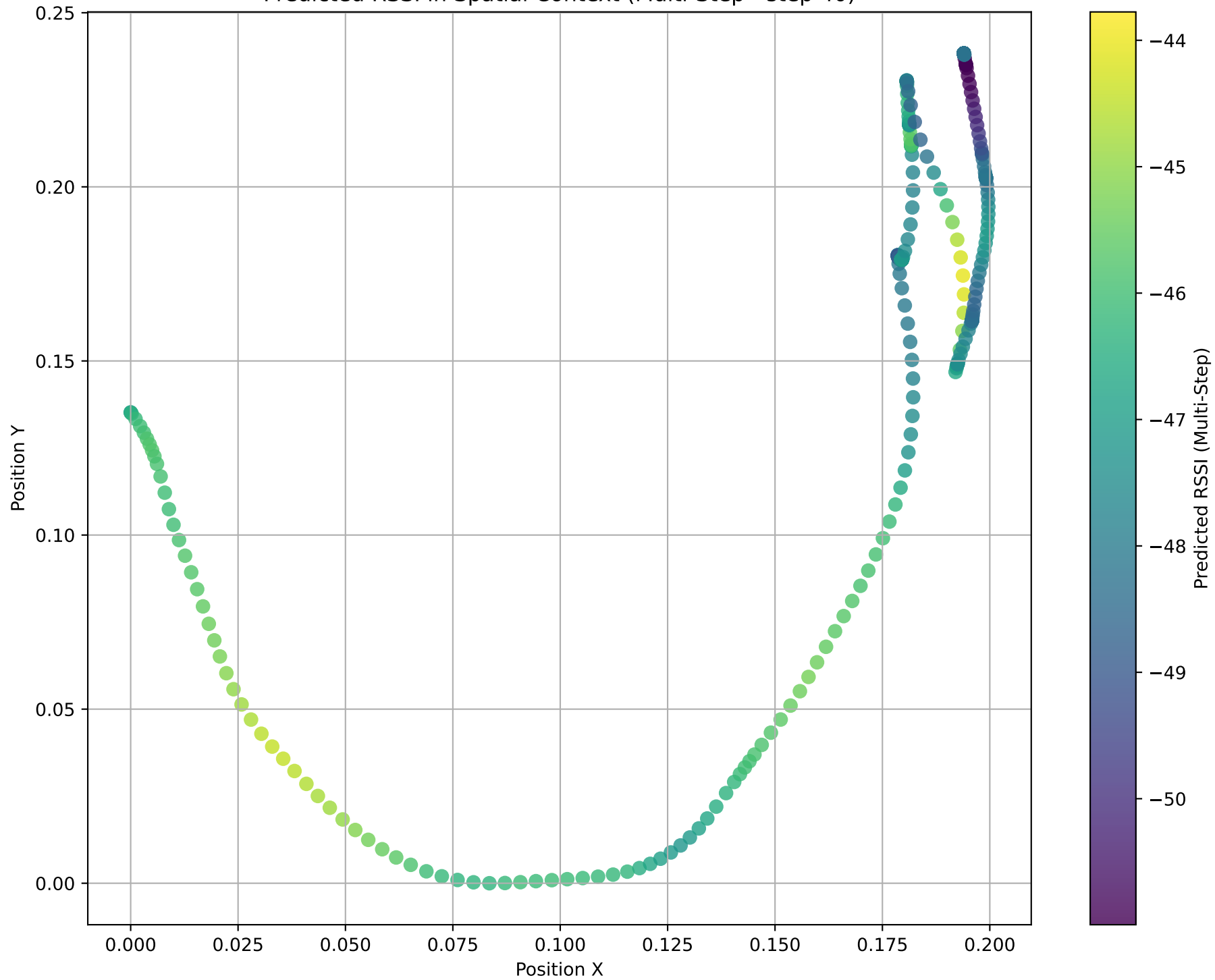
Distribution of Prediction Errors (Last Step - step 40)



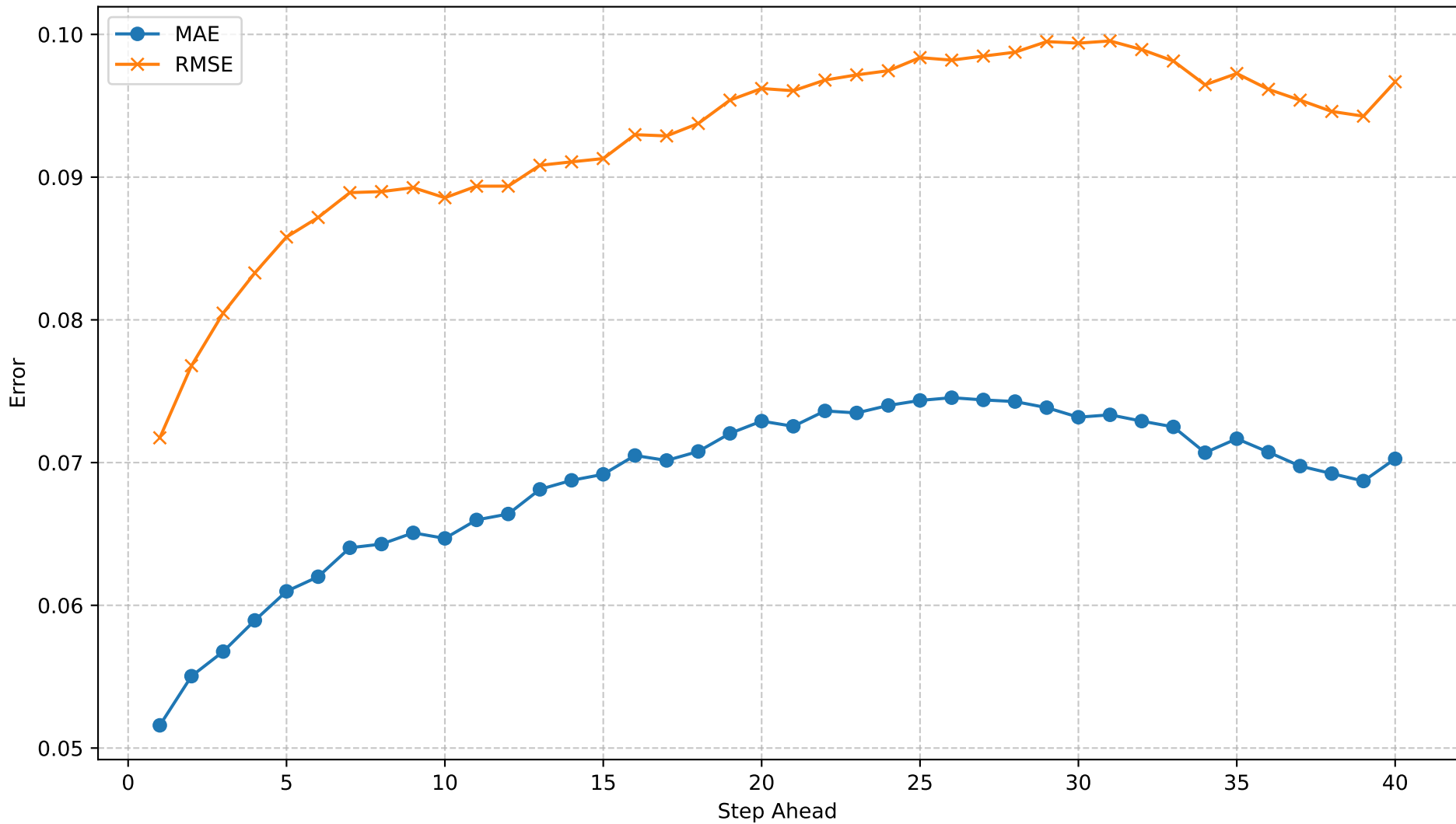
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 40)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset-ORIGINAL.parquet

Modelo: LSTM

Forecast steps: 40

MAE / RMSE (dB)

- 1º passo : 0.05 / 0.07
- Mediana : 0.07 / 0.09
- 40º passo : 0.07 / 0.10

Crescimento do erro: 36.2% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 26 (0.07), 27 (0.07), 25 (0.07)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.